

Endpoint saddles and the local classification of eigenvalue–torsion critical domains

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Abstract

We study the scale-invariant products of the first Dirichlet eigenvalue and a positive power of torsional rigidity, normalized by either volume or perimeter. At each upper local stability threshold identified by Amato, Nitsch, Trombetti and Villone, we obtain a nonlinear saddle among smooth strictly convex domains. We also classify every sufficiently near-ball critical domain, up to similarities, into finitely many analytic branches indexed by the multiplicities of two eigenvalues of a trace-free symmetric matrix. Unbalanced branches cross the threshold. Balanced branches have nonzero quadratic parameter variation; in the plane the unique nonball branch lies above the threshold. Consequently the ball is isolated among critical domains at the endpoint despite being a saddle, and an interval of strict local but nonglobal ball maximality follows. The argument combines explicit third- and fourth-order PDE expansions, an analytic boundary equation, and a finite-dimensional symmetry reduction. A classical transcendence theorem for rational-order Bessel zeros excludes exceptional dimensions. The global convex perimeter maximization problem remains unresolved.

1. Introduction and main results

For a bounded domain $\Omega \subset \mathbb{R}^n$, let $\lambda_1(\Omega)$ be the first Dirichlet eigenvalue of $-\Delta$, and let $T(\Omega) = \int_{\Omega} w$ where $-\Delta w = 1$ in Ω and $w = 0$ on its boundary. Let $V(\Omega) = |\Omega|$ and $P(\Omega) = \mathcal{H}^{n-1}(\partial\Omega)$. We consider

$$F_q(\Omega) = \frac{\lambda_1(\Omega)T(\Omega)^q}{V(\Omega)^{\alpha_q}}, \quad \alpha_q = \frac{(n+2)q-2}{n}, \quad G_q(\Omega) = \frac{\lambda_1(\Omega)T(\Omega)^q}{P(\Omega)^{\beta_q}}, \quad \beta_q = \frac{(n+2)q-2}{n-1}. \quad (1.1)$$

Both are invariant under translations and positive dilations. We write $B = B_1$, $\nu = n/2 - 1$, $j = j_{\nu,1}$, $s = j^2$ and $x = s/n$. Their upper local stability thresholds are

$$q_V = \frac{2(x-1)}{n+2}, \quad q_P = \frac{4(n-1)x-2n+6}{(n+2)(3n-1)}. \quad (1.2)$$

These thresholds and their Hessians are established in [Amato-Nitsch-Trombetti-Villone](#), Theorems 1.6 and 6.3 and Corollaries 5.14 and 6.7. At each threshold the Hessian is negative off the degree-two harmonics, after removing translations and dilations, and vanishes on that finite-dimensional kernel. Their Remark 5.15 leaves the nonlinear volume endpoint undecided. For parameters strictly above the thresholds, their results give strict local ball maximality in the stated smooth topology. We use this established spectral and stability information; the results below concern the nonlinear endpoint and the complete nearby critical set.

Theorem A (endpoint saddles). In every dimension $n \geq 2$, the ball is a saddle for F_{q_V} and G_{q_P} , even among smooth strictly convex domains. The first assertion holds at exactly fixed volume and the second at exactly fixed perimeter. In dimensions at least three the two signs occur along opposite directions of a degree-two deformation. In the plane they occur along two paths with the same first velocity and different second-order corrections.

Theorem B (local classification). Fix $n \geq 2$, an integer $k \geq 4$ and $0 < \gamma < 1$. For either family, near its parameter q_* and near the ball in $C^{k,\gamma}$, all unrestricted critical domains, up to similarities, are the ball and the analytic branches

$$R_{p,t}(\omega) = 1 + tH_p(\omega) + O(t^2), \quad H_p(\omega) = \sum_{i=1}^p \omega_i^2 - \frac{p}{n}, \quad 1 \leq p \leq \lfloor n/2 \rfloor. \quad (1.3)$$

The domains on the p th branch have $O(p) \times O(n-p)$ symmetry and are strictly convex. With $m_{r,p} = \langle H_p^r \rangle$, their parameter slopes for $p < n/2$ are

$$q'_p(0) = -\frac{3C_* m_{3,p}}{2 \partial_q e_2 m_{2,p}} \neq 0, \quad (1.4)$$

where the explicitly computed cubic coefficients C_* and the known Hessian multipliers e_2 are given below. In even dimensions the balanced branch $p = n/2$ instead satisfies $q_p(t) = q_* + \eta_n t^2 + O(t^4)$ with an explicit nonzero η_n . The opposite signs of t on that balanced branch are congruent. In dimension two, $\eta_2 > 0$ for both families.

Corollary C (isolated endpoint criticality). At $q = q_*$ the ball is the only critical domain in a sufficiently small neighborhood, up to similarities. In the plane, for each $q > q_*$ sufficiently close to q_* there is exactly one nearby nonball similarity class, and for $q \leq q_*$ sufficiently close there is none. In odd dimensions there are $(n-1)/2$ nonball classes for either sign of sufficiently small $q - q_*$. In even dimensions at least four, the $n/2 - 1$ unbalanced classes occur on both sides and the balanced class occurs on the side determined by η_n . A common sign for η_n in every higher even dimension is not asserted.

Corollary D (separation of local and global maximality). For each family and dimension there is $\varepsilon_n > 0$ such that, for $q_* < q < q_* + \varepsilon_n$, the ball is a strict local maximizer but not a global maximizer, even among smooth strictly convex domains. No explicit value of ε_n is claimed.

The volume branch lies in a different sign regime from the rigidity theorem of [Celentano-Nitsch-Trombetti](#), whose Corollary 1.3 assumes nonpositive product of the exponents in $T^a \lambda_1^b$. Here both exponents are positive. The argument does not contradict that result.

The general implicit-function and symmetry methods are classical. The background includes Crandall-Rabinowitz's simple-kernel bifurcation theory and the literature on spherical symmetry, including [Chossat-Lauterbach-Melbourne](#). We supply the reduction rather than invoke a new general bifurcation principle or claim a new general classification of equivariant equations. The shape-dependent content is the explicit nonlinear expansion, the verification of the analytic boundary map and its complement inverse, and the nonvanishing of the coefficients in every dimension. The latter uses the classical Siegel theorem as stated by [Lorch-Muldoon](#): a nonzero zero of a Bessel function of rational order is transcendental.

The recent preprint of [Masiello-Paoli-Salerno](#), posted September 15, 2026, concerns saddle shapes for k -Hessian torsion under quermassintegral constraints and for curvature integrals. Its Theorems 1.1-1.4 concern different functionals and a second-order saddle mechanism. No claim that ball saddles or shape bifurcations are new phenomena in general is made here.

The perimeter problem at $q = 1$, including the planar conjecture $\lambda_1(\Omega)T(\Omega)/P(\Omega)^2 \leq j_{0,1}^2/(32\pi)$ for convex Ω , is outside our conclusion. Local endpoint instability and critical-domain classification do not settle that global inequality.

2. Analytic boundary equation and Hessian

2.1. The analytic boundary equation

Let $R = 1 + h$. Write u for the positive first Dirichlet eigenfunction normalized by $\int_{\Omega_h} u^2 = 1$, w for the torsion function, and κ for the sum of the outward principal curvatures. Put $E_F = \log F_q$ and $E_G = \log G_q$. The Hadamard formulas give their shape densities

$$\begin{aligned}\psi_F &= -\frac{u_\nu^2}{\lambda_1} + q \frac{w_\nu^2}{T} - \frac{\alpha_q}{V}, \\ \psi_G &= -\frac{u_\nu^2}{\lambda_1} + q \frac{w_\nu^2}{T} - \frac{\beta_q}{P} \kappa.\end{aligned}\tag{2.1}$$

Normal derivatives are outward; only their squares enter. For a radial variation v , the product of its normal speed and the surface Jacobian is vR^{n-1} . Thus, with normalized spherical average as inner product,

$$D_h E(q, h)[v] = \langle Z(q, h), v \rangle, \quad Z = |S^{n-1}| R^{n-1} \psi.\tag{2.2}$$

The PDE normalizations in (2.1) matter: the eigenvalue term uses an L^2 -normalized eigenfunction. First-variation formulas are also stated in Theorems 2.7-2.8 of Celentano-Nitsch-Trombetti, [final 2024 article](#).

Let $K = O(n-1) \times \{\pm 1\}$ act by rotations on the last $n-1$ coordinates and reversal of the first. For $n=2$ this is reflection in both coordinate axes. Set

$$\begin{aligned}X &= \{h \in C^{k,\gamma}(S^{n-1})^K : \langle h \rangle = 0\}, \\ Y_F &= C^{k-1,\gamma}(S^{n-1})^K \cap \{\langle f \rangle = 0\}, \\ Y_G &= C^{k-2,\gamma}(S^{n-1})^K \cap \{\langle f \rangle = 0\}.\end{aligned}\tag{2.3}$$

Define $\Psi(q, h) = Z(q, h) - \langle Z(q, h) \rangle$, using the appropriate Y . These are real-analytic maps on an open neighborhood of $(q_*, 0)$ in $\mathbb{R} \times X$.

Here are the analytic details. Choose a fixed smooth radial cutoff χ equal to zero near the origin and to one near $r=1$. The map $r\omega \mapsto r(1 + \chi(r)h(\omega))\omega$ is a $C^{k,\gamma}$ diffeomorphism from the ball to Ω_h when h is small. Pulling the Dirichlet equations back to the fixed ball gives second-order operators whose coefficients are analytic functions of h , its first and second derivatives, and the inverse of a matrix close to the identity. They act boundedly from zero-boundary $C^{k,\gamma}$ to $C^{k-2,\gamma}$.

The Dirichlet Laplacian is invertible on these spaces by the classical Schauder existence and estimate theorem. The torsion solution therefore depends analytically on h , by the analytic implicit-function theorem. For the first eigenpair use the equation together with L^2 normalization, with the pulled-back volume density included. Its linearization in (u, λ) is invertible: projection onto the simple ground state determines λ , the normalization determines the ground-state component of u , and the Dirichlet resolvent determines its orthogonal component. The same implicit-function argument gives analytic dependence. Continuity of the spectrum and positivity select the first eigenpair. The pulled-back normal derivatives are analytic in $C^{k-1,\gamma}$, and the curvature is analytic in $C^{k-2,\gamma}$. Integration, logarithms of positive quantities, and multiplication preserve analyticity. These observations prove the mapping assertions without assuming a formal shape Taylor series converges.

The ball satisfies $\Psi(q, 0) = 0$ for every nearby q . Further, $\Psi(q, h) = 0$ is equivalent to unrestricted criticality on this symmetry class. Indeed Z is K -invariant, so $\Psi = 0$ makes it a constant. Scale invariance gives

$$0 = D_h E(q, h)[1 + h] = \langle Z, 1 + h \rangle = Z, \quad (2.4)$$

because $\langle h \rangle = 0$. Thus $Z = 0$ pointwise, and (2.2), or (2.1) for arbitrary normal speeds, yields every first-variation equation. Conversely unrestricted criticality implies $Z = 0$. There is no residual Lagrange multiplier or restriction to symmetric test variations in this conclusion.

2.2. Linearization and the complement inverse

For a spherical harmonic of degree $\ell \geq 2$, put

$$c_\ell = j \frac{J_{\nu+\ell+1}(j)}{J_{\nu+\ell}(j)}, \quad d_\ell = \ell - c_\ell. \quad (2.5)$$

The denominator cannot vanish because the first Dirichlet eigenspace is radial and simple. The normalized regular radial Helmholtz extension of a degree- ℓ boundary harmonic has outward derivative d_ℓ .

For a single harmonic H_ℓ the coefficient of $t^2 \langle H_\ell^2 \rangle$ in the logarithmic functional along $h = tH_\ell$ is

$$\begin{aligned} e_\ell^F(q) &= 2d_\ell + 2(n-1) - (n+2)q(\ell-1), \\ e_\ell^G(q) &= 2d_\ell + n-1 + \frac{(n+2)q(n+1-2\ell)}{2} \\ &\quad - \frac{\beta_q}{2} \{ \ell(\ell+n-2) + (n-1)(n-2) \}. \end{aligned} \quad (2.6)$$

For completeness, the eigenvalue's normalized quadratic coefficient is $2d_\ell + n-1$, by the second-order compatibility calculation of Section 3 with degree two replaced by ℓ . The torsion coefficient is $(n+2)((n+1)/2 - \ell)$: its first correction is $r^\ell H_\ell/n$, its second boundary value is $(1/2 - \ell)H_\ell^2/n$, and the signed shell integral contributes $|S| \langle H_\ell^2 \rangle / (2n)$. The volume coefficient is $n(n-1)/2$. The perimeter coefficient is $[\ell(\ell+n-2) + (n-1)(n-2)]/2$. Combining them proves (2.6). Polarization and orthogonality show that $L = D_h \Psi(q_*, 0)$ acts on each harmonic as multiplication by $2e_\ell(q_*)$.

The source's Corollaries 5.14 and 6.7 and their proofs give

$$e_2(q_*) = 0, \quad e_\ell(q_*) < 0 \quad (\ell \geq 3). \quad (2.7)$$

This spectral information is an established input, not a new endpoint conclusion. The K -invariant harmonics of degree two form exactly a line, spanned by $H = \omega_1^2 - 1/n$ for $n \geq 3$, or by $H = \cos 2\theta$ for $n = 2$. The invariance eliminates degree one. Thus $\ker L = \mathbb{R}H$. Directly from (2.6),

$$\partial_q e_2^F = -(n+2), \quad \partial_q e_2^G = -\frac{(n+2)(3n-1)}{2(n-1)}. \quad (2.8)$$

We also need a bounded inverse on the complement in the actual Hölder spaces, not merely nonzero formal multipliers. Let D denote the Laplace Dirichlet-to-Neumann map on the sphere. On mean-zero functions it is an isomorphism $C^{k,\gamma} \rightarrow C^{k-1,\gamma}$: the inverse is the harmonic Neumann solution with its constant fixed to have mean-zero trace. This follows from the Neumann Schauder theorem.

The difference between the Helmholtz and Laplace Dirichlet-to-Neumann maps is compact as a map from $C^{k,\gamma}$ to $C^{k-1,\gamma}$ on mean-zero traces. Indeed the harmonic extension v has zero spherical mean at every radius. The equation

$$(-\Delta - j^2)z = j^2 v, \quad z|_S = 0, \quad (2.9)$$

has a unique solution with zero spherical mean, since the only resonant eigenspace is radial. The Dirichlet Schauder estimate on this complement bounds z in $C^{k+2,\gamma}$ by the boundary trace in $C^{k,\gamma}$. The normal derivative of z lies in $C^{k+1,\gamma}$, whose inclusion in $C^{k-1,\gamma}$ is compact. The sum $v + z$ is the Helmholtz extension. Thus the operator with multipliers c_ℓ is compact at the required orders.

For F , (2.6) now expresses L as a compact perturbation of

$$2[2 - (n + 2)q_V]D : X \longrightarrow Y_F. \quad (2.10)$$

For G , it expresses L as a compact perturbation of

$$-\beta_{q_P}(-\Delta_S) : X \longrightarrow Y_G. \quad (2.11)$$

Both leading operators are invertible on mean-zero invariant functions. In fact $j^2 > 2n$ (the standard Bessel-zero bound also used in source Remark 1.7) gives $q_V, q_P > 2/(n + 2)$, so the coefficient in (2.10) is nonzero and $\beta_{q_P} > 0$. Hence L is Fredholm of index zero. Harmonic testing in (2.6) shows its range is orthogonal to H ; the codimension is exactly one by its index and kernel. If

$$X_\perp = \{h \in X : \langle hH \rangle = 0\}, \quad Y_\perp = \{f \in Y : \langle fH \rangle = 0\}, \quad (2.12)$$

then $L : X_\perp \rightarrow Y_\perp$ is a bounded isomorphism.

3. Cubic endpoint expansion in every dimension

3.1. A cubic expansion along a degree-two harmonic

Let h be a real spherical harmonic of degree two on S^{n-1} , write $\langle \cdot \rangle$ for normalized spherical average, and set

$$\Omega_t = \{r\omega : 0 \leq r < 1 + th(\omega)\}, \quad m_2 = \langle h^2 \rangle, \quad m_3 = \langle h^3 \rangle. \quad (3.1)$$

Here $\langle h \rangle = 0$ and $-\Delta_S h = 2nh$. Put

$$d = x - n, \quad a = d + \frac{n-1}{2}, \quad B = -x^2 + \frac{2n+3}{3}x - \frac{n(n-1)}{6}. \quad (3.2)$$

The necessary expansions are

$$\begin{aligned} \frac{\lambda_1(\Omega_t)}{j^2} &= 1 + 2am_2t^2 + 2Bm_3t^3 + O(t^4), \\ \frac{T(\Omega_t)}{T(B_1)} &= 1 + \frac{(n+2)(n-3)}{2}m_2t^2 + \frac{(n+2)(n^2-5n+12)}{6}m_3t^3 + O(t^4), \\ \frac{|\Omega_t|}{|B_1|} &= 1 + \frac{n(n-1)}{2}m_2t^2 + \frac{n(n-1)(n-2)}{6}m_3t^3 + O(t^4), \\ \frac{P(\Omega_t)}{P(B_1)} &= 1 + \frac{n^2-n+2}{2}m_2t^2 + \frac{(n-3)(n^2+2)}{6}m_3t^3 + O(t^4). \end{aligned} \quad (3.3)$$

The eigenvalue and torsion remainders will be justified in Section 3.2. The volume formula follows by expanding $(1 + th)^n$. For perimeter, expand

$$\frac{P(\Omega_t)}{P(B_1)} = \left\langle (1+th)^{n-1} \sqrt{1 + \frac{t^2 |\nabla_S h|^2}{(1+th)^2}} \right\rangle. \quad (3.4)$$

Use $\langle |\nabla_S h|^2 \rangle = 2nm_2$ and $\langle h|\nabla_S h|^2 \rangle = nm_3$. The latter follows by integrating $\nabla_S(h^2) \cdot \nabla_S h = -h^2 \Delta_S h$.

Taking logarithms in (3.3), no cross terms enter before order four. The quadratic coefficients of $\log(F_q(\Omega_t)/F_q(B_1))$ and $\log(G_q(\Omega_t)/G_q(B_1))$ are, respectively,

$$[2x - 2 - (n+2)q]m_2, \quad \left[2x - 1 + \frac{2}{n-1} - \frac{(n+2)(3n-1)}{2(n-1)}q \right] m_2. \quad (3.5)$$

They vanish at (1.2). At their respective critical exponents the cubic coefficients are

$$\begin{aligned} C_V(n, x) &= -2x^2 + \frac{2(n+8)}{3}x - \frac{8}{3}, \\ C_P(n, x) &= -2x^2 + \frac{2(3n^2 + 22n - 9)}{3(3n-1)}x - \frac{4(5n-3)}{3(3n-1)}. \end{aligned} \quad (3.6)$$

Thus the two logarithmic variations equal $C_V(n, x)m_3t^3 + O(t^4)$ and $C_P(n, x)m_3t^3 + O(t^4)$.

3.2. Derivation of the PDE expansions and their remainders

3.2.1. Eigenvalue

Let $u_0(r) = r^{-\nu} J_\nu(jr)$, interpreted smoothly at zero, $p = u'_0(1) \neq 0$, and $I = \int_{B_1} u_0^2$. The standard Rellich identity, obtained by testing the eigenvalue equation against $r\partial_r u_0$, gives

$$p^2 |S^{n-1}| = 2j^2 I. \quad (3.7)$$

Let H_2 be the regular radial solution of the degree-two Helmholtz equation at j^2 , normalized by $H_2(1) = 1$. Then

$$H_2(r) = \frac{r^{-\nu} J_{\nu+2}(jr)}{J_{\nu+2}(j)}, \quad H'_2(1) = d = \frac{j^2}{n} - n, \quad H''_2(1) = -(n-1)d - j^2 + 2n. \quad (3.8)$$

The recurrence $J_{\nu+2}(j) = nJ_{\nu+1}(j)/j$ and the derivative recurrence prove the middle identity. The denominator is nonzero; otherwise a degree-two eigenfunction would belong to the first eigenspace of the ball, contradicting its simplicity. The last identity is the radial ODE at $r = 1$. The radial equation also gives

$$u''_0(1) = -(n-1)p, \quad u'''_0(1) = [n(n-1) - j^2]p. \quad (3.9)$$

We construct a finite expansion $U_t = u_0 + tu_1 + t^2u_2 + t^3u_3$ and $\ell_t = j^2 + \lambda_2 t^2 + \lambda_3 t^3$. Let $L = -\Delta - j^2$. Set

$$u_1 = -pH_2h, \quad (3.10)$$

so $Lu_1 = 0$ and $u_1|_S = -ph$. At second order the equations are

$$Lu_2 = \lambda_2 u_0, \quad u_2|_S = -h\partial_r u_1 - \frac{1}{2}h^2 u''_0(1) = pah^2. \quad (3.11)$$

Green's formula for a function with boundary value g gives $\int_{B_1} u_0 Lu = \int_S pg$. Therefore the Fredholm compatibility condition for (3.11) is

$$\lambda_2 I = p^2 a |S^{n-1}| m_2, \quad \lambda_2 = 2j^2 a m_2. \quad (3.12)$$

The degree-two part of u_2 is $paH_2\Pi_2(h^2)$, where Π_2 is orthogonal projection onto degree-two harmonics. Its remaining parts have degrees zero and four. Orthogonality implies

$$\langle h\partial_r u_2 \rangle = pad m_3. \quad (3.13)$$

This is independent of the arbitrary addition of a multiple of u_0 to u_2 .

At order three choose u_3 to satisfy

$$Lu_3 = \lambda_2 u_1 + \lambda_3 u_0, \quad u_3|_S = -h\partial_r u_2 - \frac{1}{2}h^2\partial_r^2 u_1 - \frac{1}{6}h^3 u_0'''(1). \quad (3.14)$$

Since $\int u_0 u_1 = 0$, compatibility and (3.7)-(3.13) give

$$\lambda_3 = 2j^2 \left[-ad + \frac{H_2''(1)}{2} - \frac{n(n-1) - j^2}{6} \right] m_3 = 2j^2 B m_3. \quad (3.15)$$

This proves the asserted coefficient algebra without assuming that the formal series is already an eigenfunction.

Here is the remainder argument. The compatibility conditions ensure that (3.11) and (3.14) have smooth solutions on the ball. All their data involve finitely many spherical harmonics (degrees at most six); separation of variables gives regular radial ODE solutions. These extend smoothly to a fixed ball of radius $1 + \varepsilon$, because the radial ODE coefficients are smooth at $r = 1$. Thus the finite U_t is defined on Ω_t and satisfies

$$(-\Delta - \ell_t)U_t = O(t^4), \quad U_t(1 + th, \omega) = O(t^4) \quad (3.16)$$

in fixed sufficiently strong norms. The first statement follows by multiplying the finite polynomials in t and using (3.10)-(3.14); the second follows from Taylor expansion at $r = 1$ and the boundary conditions imposed at each order.

Use an angular boundary lift, supported in $r > (1 + th)/2$, of the second residual. Its H^1 norm is $O(t^4)$. Subtracting it produces $v_t \in H_0^1(\Omega_t)$ with weak eigenvalue residual $O(t^4)$. Domain inclusion between balls of radii $1 \pm C|t|$, compactness of the positive normalized first eigenfunctions, and simplicity of the disk/ball ground state show that their pairing with v_t converges to a nonzero constant. Testing the residual against that first eigenfunction gives

$$\lambda_1(\Omega_t) = \ell_t + O(t^4). \quad (3.17)$$

The compactness and pairing argument is written explicitly in Section 4.4 and is dimension-independent. This completes the eigenvalue part of (3.3).

3.2.2. Torsion

Let $\mathcal{H}$ be harmonic extension from S^{n-1} into B_1 and $D = \partial_r \mathcal{H}|_S$; D acts by multiplication by the degree on each spherical harmonic. The torsion coefficients are

$$w_0 = \frac{1-r^2}{2n}, \quad w_1 = \frac{r^2 h}{n}, \quad w_2 = -\frac{3}{2n} \mathcal{H}(h^2), \quad (3.18)$$

and w_3 is the harmonic extension of

$$w_3|_S = \frac{3}{2n} h D(h^2) - \frac{1}{n} h^3. \quad (3.19)$$

These follow by cancelling the boundary values of $w_0 + tw_1 + t^2 w_2 + t^3 w_3$ at $r = 1 + th$ order by order. The sum has Laplacian exactly -1 . Orthogonality and $Dh = 2h$ give

$$\langle h D(h^2) \rangle = 2m_3, \quad \langle w_3|_S \rangle = \frac{2}{n} m_3. \quad (3.20)$$

For any harmonic extension, its integral over the ball is $|S^{n-1}|/n$ times its boundary average. To include the moving boundary, integrate the coefficients over B_1 and expand the signed shell integrals from 1 to $1 + th$. At order two the shell contributes $|S^{n-1}|m_2/(2n)$. At order three it contributes $|S^{n-1}|(n-5)m_3/(6n)$. In detail, the three cubic shell terms, divided by $|S^{n-1}|m_3$, are

$$-\frac{2n-1}{6n} (w_0), \quad \frac{n+1}{2n} (tw_1), \quad -\frac{3}{2n} (t^2 w_2). \quad (3.21)$$

Adding the ball integrals and dividing by $T(B_1) = |S^{n-1}|/[n^2(n+2)]$ gives the second line of (3.3). All harmonic terms are finite harmonic polynomials and extend across the sphere. The uncanceled boundary value is $O(t^4)$; the maximum principle bounds the actual torsion correction by $O(t^4)$ uniformly, and hence also in its integral. This supplies the remainder claimed in (3.3).

3.3. Nonvanishing and the higher-dimensional saddle

For each fixed integer $n \geq 3$, both expressions in (3.6) are nonzero quadratic polynomials in x with rational coefficients; each has leading coefficient -2 . The classical theorem of Siegel says that every nonzero zero of J_ν is transcendental for rational ν . The applicable statement is given in the introduction of Lorch and Muldoon, *Transcendentality of zeros of higher derivatives of functions involving Bessel functions*, International Journal of Mathematics and Mathematical Sciences 18(3) (1995), 551-560, [published PDF](#), [public journal mirror](#).

Since $\nu = n/2 - 1$ is rational, j is transcendental. Consequently $x = j^2/n$ is transcendental: an algebraic value of x would make j algebraic as a root of $z^2 - nx$. Thus

$$C_V(n, x) \neq 0, \quad C_P(n, x) \neq 0. \quad (3.22)$$

No claim that these coefficients have a common sign across dimensions is needed.

Choose

$$h(\omega) = \omega_1^2 - \frac{1}{n}. \quad (3.23)$$

This is a degree-two harmonic because its homogeneous extension is $z_1^2 - |z|^2/n$. The spherical moments $\langle \omega_1^2 \rangle = 1/n$, $\langle \omega_1^4 \rangle = 3/[n(n+2)]$, and $\langle \omega_1^6 \rangle = 15/[n(n+2)(n+4)]$ give

$$m_3 = \frac{8(n-1)(n-2)}{n^3(n+2)(n+4)} > 0. \quad (3.24)$$

These moment identities follow, for example, by integrating the first-coordinate marginal with density proportional to $(1 - z^2)^{(n-3)/2}$ and applying the beta-integral recurrence. From (3.6), (3.22) and (3.24), each critical logarithmic variation has a nonzero cubic leading term. Taking t of opposite signs proves the saddle property for each functional. The smooth radial hypersurfaces are strictly convex for small $|t|$, since their second fundamental forms converge uniformly to that of the unit sphere.

For exact constraints use $[|B_1|/|\Omega_t|]^{1/n}\Omega_t$ for F and $[P(B_1)/P(\Omega_t)]^{1/(n-1)}\Omega_t$ for G . Both scalar factors tend to 1, preserve strict convexity and preserve the scale-invariant functional values. This proves the theorem for all $n \geq 3$.

4. The planar volume endpoint

4.1. Planar normalization and the two paths

Write B for the unit disk, $j = j_{0,1}$, $s = j^2$, and

$$q_* = \frac{s-2}{4}, \quad F_q(\Omega) = \frac{\lambda_1(\Omega)T(\Omega)^q}{|\Omega|^{2q-1}}. \quad (4.1)$$

Here $-\Delta w = 1$ with zero Dirichlet data and $T = \int w$. This normalization is important: the exponent in the area denominator is $2q - 1$.

Planar endpoint assertion. The disk is a saddle for F_{q_*} among smooth strictly convex planar domains, including under an exact area constraint. More precisely, set

$$\Omega_{t,b} = \{(r, \theta) : 0 \leq r < 1 + t \cos(2\theta) + bt^2 \cos(4\theta)\}, \quad b \in \{0, 1\}. \quad (4.2)$$

For all sufficiently small nonzero t ,

$$F_{q_*}(\Omega_{t,0}) > F_{q_*}(B), \quad F_{q_*}(\Omega_{t,1}) < F_{q_*}(B). \quad (4.3)$$

Both paths have the same first normal velocity $\cos(2\theta)$. Their values differ first at fourth order. Multiplication by $(\pi/|\Omega_{t,b}|)^{1/2}$ imposes exact area π and preserves the inequalities by scale invariance. These scaled paths still have the same first normal velocity. In particular, a nonpositive Hessian does not imply a local maximum at this endpoint.

The domain boundaries are positive radial graphs and converge in every fixed C^k norm to the circle. The numerator of their signed curvature is $R^2 + 2(R')^2 - RR''$, which converges uniformly to 1. Thus these are strictly convex smooth domains for small $|t|$. The unscaled radial graph perturbations also have zero mean and zero first harmonics, as in the source's local class.

4.2. Exact expansions and the signs that suffice

For each fixed real b , the expansions used below are

$$\begin{aligned}
\frac{|\Omega_{t,b}|}{\pi} &= 1 + \frac{t^2}{2} + \frac{b^2 t^4}{2}, \\
\frac{T(\Omega_{t,b})}{\pi/8} &= 1 - t^2 + \left(\frac{15}{8} + 3b - 5b^2\right)t^4 + O(t^5), \\
\sqrt{\lambda_1(\Omega_{t,b})} &= j + k_2 t^2 + k_4 t^4 + O(t^5), \quad k_2 = \frac{j(s-3)}{4}, \\
k_4 &= \frac{jK_b(s)}{256(s-6)},
\end{aligned} \tag{4.4}$$

where

$$\begin{aligned}
K_b(s) &= 16b^2 s^2 - 576b^2 s + 2688b^2 - 24bs^3 + 440bs^2 - 1792bs + 384b \\
&\quad - 7s^4 + 110s^3 - 703s^2 + 2072s - 1920.
\end{aligned} \tag{4.5}$$

All remainders are as $t \rightarrow 0$ for fixed b ; no uniform-in- b assertion is needed. Sections 4.3 and 4.4 prove the error estimates and specify a finite recurrence for checking every coefficient.

Taking logarithms in (4.4), the quadratic coefficient in $\log(F_q(\Omega_{t,b})/F_q(B))$ is

$$\frac{s-3}{2} - q - \frac{2q-1}{2} = \frac{s-2-4q}{2}. \tag{4.6}$$

It vanishes at $q = q_*$. The quartic coefficient there is

$$\begin{aligned}
128(6-s)C_b(s) &= 176b^2 s^2 - 1024b^2 s + 24bs^3 \\
&\quad - 536bs^2 + 2560bs - 1536b \\
&\quad + 7s^4 - 102s^3 + 555s^2 - 1280s + 768.
\end{aligned} \tag{4.7}$$

Consequently

$$\log \frac{F_{q_*}(\Omega_{t,b})}{F_{q_*}(B)} = C_b(s)t^4 + O(t^5). \tag{4.8}$$

In particular, with

$$\begin{aligned}
P_0(x) &= 7x^4 - 102x^3 + 555x^2 - 1280x + 768, \\
P_1(x) &= 7x^4 - 78x^3 + 195x^2 + 256x - 768,
\end{aligned} \tag{4.9}$$

we have $C_0(s) = P_0(s)/(128(6-s))$ and $C_1(s) = P_1(s)/(128(6-s))$.

Here is an exact sign proof without numerical Bessel evaluation. The Bessel power series and alternating remainder bounds give

$$J_0(\sqrt{23/4}) \geq \sum_{k=0}^5 \frac{(-23/16)^k}{(k!)^2} = \frac{17997619}{5033164800} > 0, \tag{4.10}$$

and

$$J_0(\sqrt{29/5}) \leq \sum_{k=0}^4 \frac{(-29/20)^k}{(k!)^2} = -\frac{42533}{30720000} < 0. \tag{4.11}$$

The terms after the first decrease in magnitude at these arguments, which is sufficient for the stated tail bounds. Moreover for $0 < z \leq \sqrt{29/5}$, setting $v = z^2/4$, the series for J_1 has decreasing alternating terms and yields

$$J_1(z) \geq \frac{z}{2} \left(1 - \frac{v}{2}\right) > 0. \quad (4.12)$$

Thus $J'_0 = -J_1 < 0$ on this interval. Its first positive zero satisfies

$$\frac{23}{4} < s < \frac{29}{5} < 6. \quad (4.13)$$

For $y = x - 23/4$ one computes

$$\begin{aligned} P_0(23/4 + y) &= 7y^4 + 59y^3 + \frac{1473}{8}y^2 + \frac{4935}{16}y + \frac{4719}{256}, \\ P_1(23/4 + y) &= 7y^4 + 83y^3 + \frac{1905}{8}y^2 + \frac{1359}{16}y - \frac{6513}{256}. \end{aligned} \quad (4.14)$$

The first is positive for $y \geq 0$. The second is increasing there, and its value at $x = 29/5$ is $-12868/625 < 0$. Equations (4.13)-(4.14) prove $C_0(s) > 0 > C_1(s)$. Formula (4.8) therefore proves (4.3).

4.3. Torsion expansion with a controlled remainder

Let $R = 1 + t \cos 2\theta + bt^2 \cos 4\theta$ and use ρ as an interior radial coordinate. Seek

$$W_t(\rho, \theta) = \frac{1 - \rho^2}{4} + c_0(t) + \sum_{n \in \{2, 4, 6, 8\}} c_n(t) \rho^n \cos(n\theta). \quad (4.15)$$

This function satisfies $-\Delta W_t = 1$ exactly. The following coefficients cancel its boundary values through degree four in t :

$$\begin{aligned} c_0(t) &= -\frac{3}{8}t^2 - \frac{14b^2 - 16b - 3}{16}t^4, \\ c_2(t) &= \frac{1}{2}t - \frac{10b - 3}{8}t^3, \\ c_4(t) &= \frac{4b - 3}{8}t^2 + \frac{3(8b - 3)}{8}t^4, \\ c_6(t) &= -\frac{5(2b - 1)}{8}t^3, \\ c_8(t) &= -\frac{7(2b^2 - 8b + 3)}{16}t^4. \end{aligned} \quad (4.16)$$

These identities can be checked by substituting $\rho = R$ in (4.15), expanding powers of R , and using $2 \cos a \cos b = \cos(a+b) + \cos(a-b)$. Since only finite polynomials are involved, the uncanceled boundary value is $O(t^5)$ uniformly in θ . The actual torsion function $w_t - W_t$ is harmonic with boundary data $-W_t$. The maximum principle gives $\|w_t - W_t\|_\infty = O(t^5)$. The areas are uniformly bounded, so their integrals differ by $O(t^5)$.

Writing $\langle f \rangle = (2\pi)^{-1} \int_0^{2\pi} f d\theta$, direct integration of (4.15) gives

$$\frac{1}{2\pi} \int_{\Omega_{t,b}} W_t = \left\langle \frac{R^2}{8} - \frac{R^4}{16} + \sum_{n \in \{0,2,4,6,8\}} \frac{c_n(t)}{n+2} R^{n+2} \cos(n\theta) \right\rangle. \quad (4.17)$$

Inserting (4.16) and keeping terms through degree four proves the torsion line of (4.4). The area identity follows directly from $|\Omega| = \pi \langle R^2 \rangle$.

4.4. Eigenvalue expansion with a controlled remainder

Set $k(t) = j + k_2 t^2 + k_4 t^4$ and consider the finite Bessel sum

$$\begin{aligned} U_t(\rho, \theta) = & J_0(k(t)\rho) + (A_{21}t + A_{23}t^3)J_2(k(t)\rho) \cos 2\theta \\ & + (A_{42}t^2 + A_{44}t^4)J_4(k(t)\rho) \cos 4\theta \\ & + A_{63}t^3J_6(k(t)\rho) \cos 6\theta + A_{84}t^4J_8(k(t)\rho) \cos 8\theta. \end{aligned} \quad (4.18)$$

It is smooth at the origin and satisfies $-\Delta U_t = k(t)^2 U_t$ exactly. Put

$$\begin{aligned} A_{21} &= s/2, \\ A_{42} &= -\frac{s^2(4b-s+3)}{32(s-6)}, \\ A_{23} &= -\frac{s(20bs^2 - 288bs + 960b + 7s^3 - 89s^2 + 416s - 768)}{128(s-6)}. \end{aligned} \quad (4.19)$$

The remaining coefficients A_{63}, A_{44}, A_{84} are uniquely specified by cancelling the degree-three mode 6 and degree-four modes 4 and 8 in $U_t(R, \theta)$. Their explicit expanded rational expressions are unnecessary for the constant equation. Existence here does not assume a solution of an infinite system: each coefficient first enters its indicated harmonic multiplied by $J_n(j) \neq 0$, and there are just these three scalar equations.

For completeness, nonvanishing follows from the Bessel series: for every integer $n \geq 1$ and $0 < z \leq \sqrt{29/5}$,

$$J_n(z) \geq \frac{(z/2)^n}{n!} \left(1 - \frac{z^2}{4(n+1)} \right) > 0. \quad (4.20)$$

The alternating terms decrease, since their successive magnitude ratio is $(z^2/4)/((k+1)(n+k+1)) < 1$. Here is a reproducible finite coefficient calculation. Normalize $B_n = J_n(j)/J_1(j)$, with

$$B_0 = 0, \quad B_1 = 1, \quad B_{n+1} = \frac{2n}{j} B_n - B_{n-1}, \quad J_n^{(d)}(j) = 2^{-d} \sum_{a=0}^d (-1)^a \binom{d}{a} J_{n-d+2a}(j), \quad (4.21)$$

using $J_{-n} = (-1)^n J_n$. Expand the argument increment as

$$k(t)R - j = jtc + (k_2 + jbd)t^2 + k_2 ct^3 + (k_4 + k_2 bd)t^4 + O(t^5), \quad c = \cos 2\theta, \quad d = \cos 4\theta. \quad (4.22)$$

At order one, mode 2 yields A_{21} . At order two, modes 0 and 4 yield k_2, A_{42} . At order three, modes 2 and 6 yield A_{23}, A_{63} . At order four, modes 0, 4 and 8 yield k_4, A_{44}, A_{84} . No other modes occur at these orders. In particular, the mean equation at order two is

$$-J_1(j)k_2 + \frac{jJ_1(j)}{4} + \frac{A_{21}jJ_2'(j)}{2} = 0, \quad (4.23)$$

which gives (4.4). The mean equation at order four is

$$\begin{aligned} 0 = & -J_1(j)k_4 + \frac{J_0''(j)}{2} \left(k_2^2 + \frac{j^2 b^2}{2} + jk_2 \right) + J_0'''(j) \left(\frac{j^2 k_2}{4} + \frac{j^3 b}{8} \right) + \frac{j^4 J_0''''(j)}{64} \\ & + A_{21} \left(\frac{k_2 J_2'(j)}{2} + J_2''(j) \left(\frac{jk_2}{2} + \frac{j^2 b}{4} \right) + \frac{j^3 J_2'''(j)}{16} \right) + \frac{A_{23}jJ_2'(j)}{2} \\ & + A_{42} \left(\frac{jbJ_4'(j)}{2} + \frac{j^2 J_4''(j)}{8} \right). \end{aligned} \quad (4.24)$$

It uses $\langle c^2 \rangle = 1/2$, $\langle c^4 \rangle = 3/8$, and $\langle c^2 d \rangle = 1/4$. Substitution of (4.19) and (4.21) gives exactly (4.5). This displays the scalar calculation determining the decisive fourth-order coefficient, rather than relying on a numerical eigensolver.

After all these choices, $g_t(\theta) = U_t(R, \theta) = O(t^5)$ in C^1 (indeed in every fixed C^k norm), by Taylor's theorem for finitely many smooth functions. Choose a smooth fixed cutoff χ that vanishes for arguments at most $1/2$ and equals 1 near 1. The function

$$E_t(\rho, \theta) = \chi(\rho/R(\theta))g_t(\theta) \quad (4.25)$$

has trace g_t and $H^1(\Omega_{t,b})$ norm $O(t^5)$, since R and its derivatives are uniformly controlled and R stays bounded away from zero. Thus $v_t = U_t - E_t \in H_0^1(\Omega_{t,b})$, and for every $\phi \in H_0^1(\Omega_{t,b})$,

$$\left| \int \nabla v_t \cdot \nabla \phi - k(t)^2 \int v_t \phi \right| \leq C|t|^5 \|\phi\|_{H^1}. \quad (4.26)$$

Let ϕ_t be the positive L^2 -normalized first eigenfunction. Inclusion between disks of radii $1 \pm C|t|$ gives $\lambda_1(\Omega_{t,b}) \rightarrow j^2$ and a uniform H^1 bound on ϕ_t . Their zero extensions are bounded in $H_0^1(B_2)$; compactness gives strong L^2 subsequential convergence. Every limit vanishes outside B , has norm 1, belongs to $H_0^1(B)$, and has energy at most j^2 . The Rayleigh principle and positivity identify it uniquely as the normalized first disk eigenfunction. Hence the whole family converges to that eigenfunction. Also v_t , extended by zero, converges in L^2 to $J_0(j\rho)$ on B and zero outside. It follows that $\int v_t \phi_t$ converges to a positive constant.

Taking $\phi = \phi_t$ in (4.26) therefore gives

$$|\lambda_1(\Omega_{t,b}) - k(t)^2| = O(t^5). \quad (4.27)$$

This proves the eigenvalue line of (4.4), including the remainder. Together with Sections 4.2 and 4.3 it completes the proof of (4.3).

5. The planar perimeter endpoint

For $n = 2$, every degree-two harmonic has zero cubic moment, so Section 3.3 does not settle the endpoint. Section 4 already proves the volume endpoint by a fourth-order calculation. The same paths also settle the perimeter endpoint.

Write $s = j_{0,1}^2$, $q_P = (s+1)/10$, and use $\Omega_{t,b}$ from Section 4, with radial function $R = 1 + t \cos 2\theta + bt^2 \cos 4\theta$. Direct expansion of $\langle \sqrt{R^2 + (R')^2} \rangle$ gives

$$\frac{P(\Omega_{t,b})}{2\pi} = 1 + t^2 + \left(4b^2 - \frac{3}{2}b - \frac{1}{2}\right)t^4 + O(t^5). \quad (5.1)$$

Combine this with the eigenvalue and torsion expansions proved in Section 4. The quadratic coefficient of $\log(G_q(\Omega_{t,b})/G_q(B_1))$ is $(s + 1 - 10q)/2$, which vanishes at q_P . At that exponent the quartic coefficient is

$$\begin{aligned} 640(6 - s)D_b(s) = & 1264b^2s^2 - 8960b^2s + 9216b^2 \\ & + 120bs^3 - 2776bs^2 + 13760bs - 9984b \\ & + 35s^4 - 510s^3 + 2691s^2 - 5560s + 1824. \end{aligned} \quad (5.2)$$

For $b = 0, 1$ the numerator becomes, respectively,

$$\begin{aligned} Q_0(s) &= 35s^4 - 510s^3 + 2691s^2 - 5560s + 1824, \\ Q_1(s) &= 35s^4 - 390s^3 + 1179s^2 - 760s + 1056. \end{aligned} \quad (5.3)$$

Section 4 gives the exact bracket $23/4 < s < 29/5 < 6$. Setting $y = s - 23/4$,

$$\begin{aligned} Q_0(23/4 + y) &= 35y^4 + 295y^3 + \frac{6693}{8}y^2 + \frac{22659}{16}y + \frac{33003}{256}, \\ Q_1(23/4 + y) &= 35y^4 + 415y^3 + \frac{11157}{8}y^2 + \frac{11691}{16}y - \frac{55413}{256}. \end{aligned} \quad (5.4)$$

The first polynomial is positive for $y \geq 0$. The second is increasing there and is negative even at the upper endpoint, where $Q_1(29/5) = -22048/125$. Thus $D_0(s) > 0 > D_1(s)$. The remainder is $O(t^5)$ by the PDE estimates in Section 4 and Taylor's theorem for the perimeter integrand. These opposite quartic signs prove the planar perimeter saddle. Rescaling by $2\pi/P(\Omega_{t,b})$ enforces exact perimeter without changing G_q . This completes Theorem A in every dimension.

6. Local bifurcation and branch directions

The planar branch directions proved in this section are

$$q_F(t) = q_V + c_V t^2 + O(t^4), \quad q_G(t) = q_P + \frac{2}{5}c_P t^2 + O(t^4), \quad c_V > 0, \quad c_P > 0. \quad (6.1)$$

6.1. Solving the nonlinear equation

Let $m_2 = \langle H^2 \rangle$ and let $\Pi f = \langle fH \rangle H / m_2$. Write $h = tH + w$ with $w \in X_\perp$. The analytic implicit-function theorem applied to

$$(I - \Pi)\Psi(q, tH + w) = 0 \quad (6.2)$$

gives a unique analytic solution $w = w(q, t)$ near $(q_*, 0)$. Since the ball solves the equation for all q , uniqueness gives $w(q, 0) = 0$. Moreover $D_h\Psi(q, 0)H$ is a multiple of H for every q . Differentiating (6.2) at $t = 0$ therefore gives $w_t(q, 0) = 0$, so $w = O(t^2)$ uniformly for nearby q .

The remaining scalar equation is

$$\phi(q, t) = \frac{\langle \Psi(q, tH + w(q, t)), H \rangle}{m_2} = 0. \quad (6.3)$$

Since $\phi(q, 0) = 0$, the function

$$\tilde{\phi}(q, t) = \int_0^1 \partial_t \phi(q, \tau t) d\tau \quad (6.4)$$

is analytic and satisfies $\phi = t\tilde{\phi}$. At $(q_*, 0)$ it vanishes, and

$$\partial_q \tilde{\phi}(q_*, 0) = 2\partial_q e_2(q_*) \neq 0. \quad (6.5)$$

The scalar implicit-function theorem gives a unique analytic $q = q(t)$, and hence $h(t) = tH + w(q(t), t)$. Together with the ball branch, these exhaust the solutions in the indicated symmetry class near $(q_*, 0)$. Equation (2.4) proves unrestricted criticality. Since $h(t)/t \rightarrow H \neq 0$ and its mean is zero, the domains are not centered balls. Their symmetry forces the center of any ball to be the origin, so they are not translated balls either. Strict convexity follows from convergence in C^2 to the sphere.

6.2. Direction of the branches for $n \geq 3$

Let $C_* = C_V(n, x)$ or $C_P(n, x)$ from Section 3. On the complement solution (6.2), define the reduced energy $\mathcal{E}(q, t) = E(q, tH + w(q, t)) - E(q, 0)$. Because $LH = 0$ at $q = q_*$ and $w = O(t^2)$, its cubic coefficient at the critical parameter is unchanged by eliminating w . More explicitly the quadratic form has no H - w cross term for any q , by (2.6), and the w - w term has order four. Therefore

$$\mathcal{E}(q, t) = e_2(q)m_2t^2 + C_*m_3t^3 + O(t^4 + |q - q_*||t|^3), \quad m_3 = \langle H^3 \rangle. \quad (6.6)$$

The range equation and $\langle H, w_t \rangle = 0$ imply $\partial_t \mathcal{E} = m_2\phi$. Differentiating (6.6) and inserting $q(t)$ gives

$$q'(0) = -\frac{3C_*m_3}{2\partial_q e_2 m_2}. \quad (6.7)$$

For the chosen H , Section 3 and the fourth spherical moment give

$$m_2 = \frac{2(n-1)}{n^2(n+2)}, \quad m_3 = \frac{8(n-1)(n-2)}{n^3(n+2)(n+4)} > 0. \quad (6.8)$$

The two cubic coefficients are nonzero by the classical transcendence theorem explained in Section 3. Consequently $q'(0) \neq 0$. In explicit form,

$$q'_F(0) = \frac{3C_V m_3}{2(n+2)m_2}, \quad q'_G(0) = \frac{3(n-1)C_P m_3}{(n+2)(3n-1)m_2}. \quad (6.9)$$

Thus the nonspherical curves cross the threshold rather than only touching it. Nonvanishing uses a theorem for every rational Bessel order; it is not extrapolated from a finite-dimensional computation.

6.3. The planar fourth-order reduction

Now $H = \cos 2\theta$ and $m_2 = 1/2$. Rotation through $\pi/2$ preserves the reflection-invariant spaces and sends H to $-H$. Uniqueness in the preceding reduction gives $q(-t) = q(t)$ and relates $h(-t)$ to the rotated $h(t)$. In particular $q'(0) = 0$.

At the critical parameter the quadratic term in $w(q_*, t)$ contains only the mode $\cos 4\theta$. To justify the mode selection, the second derivative of the full rotation-equivariant boundary map is bilinear. Applied twice to $\cos 2\theta$, its complex Fourier weights can only be $0, 4, -4$. Projection onto mean-zero functions

eliminates weight zero, and reflection symmetry eliminates the sine term. The inverse complement linearization preserves the remaining mode. Thus

$$w(q_*, t) = b_* t^2 \cos 4\theta + O(t^3). \quad (6.10)$$

Let $C_b(s)$ be the quartic polynomial in b in Section 4, and $D_b(s)$ the one in Section 5, with $s = j_{0,1}^2$. The fourth-order coefficient of the reduced energy is respectively C_{b_*} or D_{b_*} . An $O(t^3)$ remainder in (6.10) cannot change this coefficient: in the quadratic Taylor term it would pair with H , which is in the Hessian kernel, or enter at order five; the cubic and higher Taylor terms likewise first change at order five.

The mode-four component of (6.2) at order t^2 is exactly the derivative of this quartic polynomial with respect to b . For example $\partial_b E(q_*, tH + bt^2 \cos 4\theta) = t^2 D_b E[\cos 4\theta]$; dividing its leading order t^4 identifies the equation. Therefore

$$b_F = \operatorname{argmax}_{b \in \mathbb{R}} C_b(s), \quad b_G = \operatorname{argmax}_{b \in \mathbb{R}} D_b(s), \quad c_V = C_{b_F}(s), \quad c_P = D_{b_G}(s). \quad (6.11)$$

These maximizers are unique. The b^2 coefficients are $e_4(q_*) \langle \cos^2 4\theta \rangle < 0$ by (2.7). Equivalently they can be read directly from the displayed polynomials. The strict positivity proved in Sections 4 and 5 at $b = 0$ gives

$$c_V \geq C_0(s) > 0, \quad c_P \geq D_0(s) > 0. \quad (6.12)$$

For a fully explicit specification,

$$\begin{aligned} b_F &= -\frac{3s^3 - 67s^2 + 320s - 192}{4s(11s - 64)}, \\ b_G &= -\frac{15s^3 - 347s^2 + 1720s - 1248}{4(79s^2 - 560s + 576)}. \end{aligned} \quad (6.13)$$

The denominators are nonzero by the strictly negative degree-four Hessian. From (2.6) the planar quadratic terms are $-2(q - q_V)t^2$ and $-5(q - q_P)t^2$. The reduced critical equation consequently reads

$$-4(q - q_V)t + 4c_V t^3 + O(t^5) = 0, \quad -10(q - q_P)t + 4c_P t^3 + O(t^5) = 0 \quad (6.14)$$

along the corresponding even analytic branch. In writing the remainders we used $q(t) - q_* = O(t^2)$ and rotation symmetry. Dividing by t gives precisely (6.1). The planar branches lie strictly on the $q > q_*$ side for small nonzero t .

7. Classification without a symmetry assumption

7.1. A slice for translations and dilations

Write $\langle \cdot \rangle$ for normalized spherical average. Normalize a near-ball domain by moving its centroid to zero and then making the mean of its radial function equal to one. These operations preserve criticality because both functionals are translation and scale invariant. A sufficiently small $C^{k,\gamma}$ perturbation of the sphere remains star-shaped after this centering. The new radial function is still $C^{k,\gamma}$ and close to one: transversality of rays to the nearby boundary and the implicit-function theorem in local sphere charts give this assertion. No differentiability in a translation parameter as an operator on the same Hölder space is needed here.

Define

$$\mathcal{X} = \{\zeta \in C^{k,\gamma}(S^{n-1}) : \langle \zeta \rangle = 0, \langle \zeta \omega \rangle = 0\}. \quad (7.1)$$

For ζ small, the centroid condition can be imposed by an analytic vector $a = a(\zeta) \in \mathbb{R}^n$:

$$R(\zeta) = 1 + \zeta + a(\zeta) \cdot \omega, \quad \langle R(\zeta)^{n+1} \omega \rangle = 0. \quad (7.2)$$

At $(\zeta, a) = (0, 0)$ the derivative of the second expression with respect to a is $(n+1)I/n$. Its derivative in $\zeta \in \mathcal{X}$ is zero. The finite-dimensional implicit-function theorem with Banach parameters yields a unique analytic a , with $a(0) = 0$ and $Da(0) = 0$. The construction is $O(n)$ -equivariant. If ζ is centrally even, then $a(\zeta) = 0$ by uniqueness.

Set $\mathcal{E}(q, \zeta) = \log F_q(\Omega_{R(\zeta)})$ or $\log G_q(\Omega_{R(\zeta)})$. The analytic construction in Section 2.1 applies without a symmetry restriction. Its gradient on the slice is an analytic map to the corresponding space $\mathcal{Y} = C^{k-r,\gamma} \cap \{\ell = 0, 1 \text{ modes vanish}\}$, where $r = 1$ for F and $r = 2$ for G .

To see the gradient assertion explicitly, let Z be the radial gradient in (2.2). Differentiating (7.2) expresses $Da(\zeta)[v]$ as the inverse of the finite matrix

$$(n+1)\langle R^n \omega \otimes \omega \rangle \quad (7.3)$$

applied to $-(n+1)\langle R^n v \omega \rangle$. Thus $D_\zeta \mathcal{E}[v] = \langle Z, v + Da[v] \cdot \omega \rangle$ is represented by Z plus a finite-rank analytic correction with $C^{k,\gamma}$ coefficients. Projection away from degrees zero and one gives the claimed analytic gradient $\mathcal{G}$ with no loss beyond r derivatives.

Zero gradient on this slice is equivalent to unrestricted criticality. Indeed annihilating the tangent space to the two constraints means that the radial derivative is a linear combination of the derivative of mean radius and of the centroid moment $M(\Omega) = \int_\Omega x \, dx$. At a centered normalized domain, dilation has mean-radius derivative one and moment derivative $(n+1)M = 0$. Scale invariance therefore makes the mean-radius multiplier zero. Translation by a vector v has moment derivative $|\Omega|v$. Translation invariance then makes every moment multiplier zero. Hence the unrestricted first variation vanishes. Conversely unrestricted criticality plainly implies slice criticality.

7.2. Reduction to trace-free symmetric matrices

The linearization of $\mathcal{G}$ at the ball has precisely the harmonic multipliers from (2.6) in Section 2.2, since $Da(0) = 0$ and the ball is unrestricted critical. At $q = q_*$ its kernel is the entire space $\mathcal{H}_2$ of degree-two harmonics; every degree $\ell \geq 3$ has a strictly negative multiplier. The complement isomorphism proof in Section 2.2 works on all harmonics of degrees at least three, not just invariant ones: the same Dirichlet-to-Neumann compact perturbation, index-zero argument and harmonic testing apply.

Identify $\mathcal{H}_2$ with the space $\text{Sym}_0(n)$ of real trace-free symmetric matrices via

$$H_A(\omega) = \omega^T A \omega. \quad (7.4)$$

Let Π_2 be the harmonic projection and solve

$$(I - \Pi_2)\mathcal{G}(q, H_A + w) = 0, \quad w \perp \mathcal{H}_2. \quad (7.5)$$

The analytic implicit-function theorem supplies a unique $w = w(q, A)$, with $w(q, 0) = 0$, $D_A w(q, 0) = 0$, and $w = O(|A|^2)$. Uniqueness makes w equivariant under the action of $O(n)$. In particular, the transformation $-I$ fixes every A , so $w(q, A)$ is centrally even. Consequently the centroid correction

$a(H_A + w)$ vanishes identically on the reduced family. The slice used to remove translations therefore does not alter any coefficient computed on the even paths in the preceding calculations.

Define the reduced energy

$$\mathcal{E}(q, A) = \mathcal{E}(q, H_A + w(q, A)) - \mathcal{E}(q, 0). \quad (7.6)$$

Its critical points correspond exactly to the unrestricted critical domains near the ball. The implication follows by differentiating (7.6): the complement gradient vanishes by (7.5), and $D_A w$ is orthogonal to $\mathcal{H}_2$. It is $O(n)$ -invariant under conjugation of A .

The spherical moments are

$$\langle H_A^2 \rangle = \frac{2 \operatorname{tr} A^2}{n(n+2)}, \quad \langle H_A^3 \rangle = \frac{8 \operatorname{tr} A^3}{n(n+2)(n+4)}. \quad (7.7)$$

They follow by diagonalizing A , expanding the powers, and using the moments of squared sphere coordinates; the terms containing $\operatorname{tr} A$ vanish. The endpoint calculations of Section 3, valid for every degree-two harmonic, and the kernel property imply

$$\begin{aligned} \mathcal{E}(q, A) &= \kappa_2 e_2(q) \operatorname{tr} A^2 + \kappa_3 C_* \operatorname{tr} A^3 + O(|A|^4 + |q - q_*||A|^3), \\ \kappa_2 &= \frac{2}{n(n+2)}, \quad \kappa_3 = \frac{8}{n(n+2)(n+4)}. \end{aligned} \quad (7.8)$$

Here C_* is $C_V(n, j^2/n)$ or $C_P(n, j^2/n)$. For $n \geq 3$ it is nonzero by the classical Bessel-zero transcendence result explained in Section 3. Equation (7.8) is an analytic Taylor expansion, so its derivatives have the corresponding differentiated orders. The cubic coefficient is unchanged by the complement correction because the Hessian vanishes on $\mathcal{H}_2$ and $w = O(|A|^2)$.

7.3. Excluding three distinct matrix eigenvalues

We give the finite-dimensional argument explicitly; no novel general invariant-theory theorem is being claimed. Extend the reduced energy to diagonal matrices of arbitrary trace by setting

$$f(q, z_1, \dots, z_n) = \mathcal{E}(q, \operatorname{diag}(z_i - \bar{z})), \quad \bar{z} = \frac{1}{n} \sum_i z_i. \quad (7.9)$$

It is analytic and symmetric in the z_i . Let $g_i = \partial_{z_i} f$. Symmetry implies that $g_i - g_j$ vanishes on $z_i = z_j$, and analytic division by that linear coordinate gives

$$g_i - g_j = (z_i - z_j) B_{ij}(q, z). \quad (7.10)$$

For trace-zero z , equation (7.8) gives

$$B_{ij} = 2\kappa_2 e_2(q) + 3\kappa_3 C_*(z_i + z_j) + R_{ij}, \quad (7.11)$$

where $R_{ij} = O(|z|^2 + |q - q_*||z|)$. These are analytic remainders, not just pointwise estimates. The quotient is symmetric in i, j and equivariant under permutations. Thus $R_{ij} - R_{i\ell}$ vanishes whenever $z_j = z_\ell$. A second analytic division yields

$$B_{ij} - B_{i\ell} = (z_j - z_\ell) [3\kappa_3 C_* + O(|z| + |q - q_*|)]. \quad (7.12)$$

The remainder order can also be read term by term from the convergent analytic expansion: the higher energy terms have degree at least four in z , or have a factor $q - q_*$ and degree at least three; one differentiation and two divisions lower these degrees by three. If desired, each analytic division is justified directly by the integral formula for the difference of an analytic function at two values of one coordinate.

At a critical trace-free matrix, $g_i = 0$ for every i . Indeed the derivative on the trace-free diagonal subspace vanishes and $\sum_i g_i = 0$ by construction. If three eigenvalues z_i, z_j, z_ℓ were distinct, (7.10) would give $B_{ij} = B_{i\ell} = 0$. Equation (7.12) would contradict $C_* \neq 0$ once (q, z) is sufficiently close to $(q_*, 0)$. Hence every nonzero critical matrix has at most two distinct eigenvalues.

This proves the necessary exclusion for $n \geq 3$. For $n = 2$ every trace-free symmetric matrix already has at most two eigenvalues, so no cubic argument is required. A nonzero trace-free matrix cannot have only one eigenvalue in any dimension.

7.4. The finite list of branches and their slopes

Every trace-free matrix with exactly two eigenvalues is orthogonally conjugate to

$$A = t \left(\text{diag}(I_p, 0) - \frac{p}{n} I_n \right) \quad (7.13)$$

for some $1 \leq p \leq n - 1$ and $t \neq 0$. Exchanging the two blocks replaces (p, t) by $(n - p, -t)$, so the list may be restricted to $p \leq \lfloor n/2 \rfloor$. Its isotropy group contains $K_p = O(p) \times O(n - p)$. Equivariance of the unique complement solution makes the full radial graph K_p -invariant.

The fixed subspace of $\mathcal{H}_2$ under K_p is exactly the line spanned by H_p . Moreover the reduced gradient at tH_p belongs to this fixed line. Thus unrestricted criticality there is equivalent to its scalar equation along t . The construction of Sections 2.2 and 6.1 applies with K replaced by K_p ; the kernel remains one-dimensional and $\partial_q e_2 \neq 0$. It gives a unique analytic curve $q_p(t)$ and no other nontrivial solution near the origin in this class. Because every critical matrix has the form (7.13), this list exhausts all nearby critical domains.

For the moments one can use the beta distribution of $\sum_{i=1}^p \omega_i^2$, whose parameters are $p/2, (n - p)/2$. It gives

$$m_{2,p} = \frac{2p(n - p)}{n^2(n + 2)}, \quad m_{3,p} = \frac{8p(n - p)(n - 2p)}{n^3(n + 2)(n + 4)}. \quad (7.14)$$

The slope formula from Section 6 is therefore

$$q'_p(0) = -\frac{3C_* m_{3,p}}{2\partial_q e_2 m_{2,p}}, \quad (7.15)$$

which is nonzero for $p < n/2$. When $p = n/2$, an orthogonal transformation exchanges the equal-sized coordinate blocks and sends H_p to $-H_p$. Uniqueness makes $q_p(t)$ even and identifies the two signs of t geometrically. Section 8 proves

$$q_p(t) = q_* + \eta_n t^2 + O(t^4), \quad \eta_n \neq 0, \quad (7.16)$$

by an explicit fourth-order calculation. For $n = 2$, $H_1 = \frac{1}{2} \cos 2\theta$, so the planar amplitude in Section 6 equals half the parameter used here. Its positive coefficients therefore yield $\eta_2 > 0$ for both functionals.

7.5. Consequences and precise limits

Using the balanced calculation in Section 8, the ball is an isolated critical domain at $q = q_*$ modulo similarities, even though it is a nonlinear saddle at that parameter. For unbalanced branches this follows from (7.15); for the balanced branch it follows from (7.16). Finitely many branches allow one common sufficiently small neighborhood.

In the plane, for every fixed $q > q_*$ sufficiently close to q_* , there is exactly one nearby nonball critical domain up to similarities. For $q \leq q_*$ sufficiently close there is none. Indeed the even analytic curve has the form $q(t) = Q(t^2)$ with $Q'(0) > 0$, so it is invertible as a function of $t^2 \geq 0$; the two signs represent rotations of the same shape.

In odd dimensions there are $(n-1)/2$ nearby nonball similarity classes at each sufficiently close $q \neq q_*$. In even dimensions at least four, the $n/2 - 1$ unbalanced classes occur on both sides, and the balanced class adds one on the side determined by the nonzero coefficient η_n . No universal sign of that balanced coefficient in dimensions at least four is asserted here. Different $p < n/2$ give distinct eigenvalue multiplicities in their degree-two projections and hence cannot be similar after normalization.

8. Balanced fourth-order expansion

8.1. The balanced harmonic

Let $n \geq 2$ be even, $s = j_{n/2-1,1}^2$, and set

$$\begin{aligned} h &= \sum_{i=1}^{n/2} \omega_i^2 - \frac{1}{2}, & Q &= h^2 - m_2, \\ m_2 = \langle h^2 \rangle &= \frac{1}{2(n+2)}, & m_4 = \langle h^4 \rangle &= \frac{3}{4(n+2)(n+6)}, \\ N = \langle Q^2 \rangle &= \frac{n}{2(n+2)^2(n+6)}. \end{aligned} \tag{8.1}$$

The beta distribution with parameters $n/4, n/4$ proves these moments. All odd moments vanish. Direct differentiation gives

$$-\Delta_S h = 2nh, \quad |\nabla_S h|^2 = 1 - 4h^2, \quad -\Delta_S Q = 4(n+2)Q. \tag{8.2}$$

Thus h and Q have harmonic degrees two and four. Consider

$$R = 1 + th + bt^2Q. \tag{8.3}$$

Block exchange sends t to $-t$, so the scalar functionals are even. We compute the critical quartic coefficients $K_V(b)$ and $K_P(b)$ of the two logarithmic functionals. Their quadratic coefficients in b are negative. Their maxima $c_V(n), c_P(n)$ are nonzero for every even n . Hence the balanced critical curves satisfy

$$q(t) = q_* + \eta_n t^2 + O(t^4), \quad \eta_n = -\frac{2c_*(n)}{\partial_q e_2 m_2} \neq 0. \tag{8.4}$$

Only nonvanishing is proved for all even dimensions; the planar sign is positive by Section 6.

8.2. Eigenvalue jet and its analytic remainder

Let f_0 be the regular radial ground state at eigenvalue s , normalized by $f_0(1) = 0, f_0'(1) = 1$. For $\ell > 0$, let f_ℓ be the regular degree- ℓ radial Helmholtz solution, normalized by $f_\ell(1) = 1$. Its boundary value is nonzero before normalization because the first eigenspace is radial. Write

$$d_2 = \frac{s}{n} - n, \quad d_4 = -n - 2 + \frac{s[n(n+2) - s]}{(n+2)[n(n+4) - 2s]}. \quad (8.5)$$

These equal $f_2'(1), f_4'(1)$ by the Bessel recurrence. Denote higher derivatives at one by $f_{\ell r}$. The radial ODE gives

$$\begin{aligned} f_{02} &= -(n-1), & f_{03} &= n(n-1) - s, \\ f_{04} &= (n-1)[2s - n(n+1)], \\ f_{22} &= -(n-1)d_2 - s + 2n, \\ f_{23} &= -(n-1)f_{22} + (3n-1-s)d_2 - 4n, \\ f_{42} &= -(n-1)d_4 - s + 4(n+2). \end{aligned} \quad (8.6)$$

Set

$$a = d_2 + \frac{n-1}{2}, \quad A = am_2, \quad a_4 = a - b, \quad (8.7)$$

$$c = -A(1 + f_{02} - d_2) - [b(f_{02} - d_2) + a_4 d_4] \frac{N}{m_2} - \left(\frac{f_{03}}{6} - \frac{f_{22}}{2} \right) \frac{m_4}{m_2}, \quad (8.8)$$

and define B by

$$\begin{aligned} -B &= \frac{f_{02}}{2}(b^2 N + A^2 + 2Am_2) + \frac{f_{03}}{2}(bN + Am_2) + \frac{f_{04}}{24}m_4 \\ &\quad - d_2 Am_2 - f_{22}(bN + Am_2) - \frac{f_{23}}{6}m_4 + a_4 \left(d_4 b + \frac{f_{42}}{2} \right) N + cd_2 m_2. \end{aligned} \quad (8.9)$$

Then

$$\log \frac{\lambda_1(\Omega_t)}{s} = 2At^2 + (2B - A^2)t^4 + O(t^6). \quad (8.10)$$

To prove this, put $\rho = (1 + At^2 + Bt^4)r$ and construct a finite sum beginning with

$$U = f_0(\rho) - tf_2(\rho)h + t^2 a_4 f_4(\rho)Q + t^3 [cf_2(\rho)h + f_6(\rho)H_6] + t^4 U_4. \quad (8.11)$$

Here H_6 is a degree-six harmonic and U_4 is a sum of degree-four and degree-eight regular Helmholtz modes. At the moving boundary,

$$\rho - 1 = th + t^2(bQ + A) + At^3 h + t^4(B + AbQ) + O(t^5). \quad (8.12)$$

The constant and degree-four equations at order two give (8.7). At order three only degrees two and six occur. Projecting onto degree two, using $\Pi_2(hQ) = (N/m_2)h$ and $\Pi_2(h^3) = (m_4/m_2)h$, gives (8.8); the remaining equation determines H_6 .

At order four, take the boundary average. The three terms in the first line of (8.9) come from f_0 . The first three terms of the second line come from $-tf_2h$. The last two come from the degree-four and

corrected degree-two terms. The degree-six term pairs with h to zero. Hence the mean equation gives (8.9). The remaining modes at this order have degrees four and eight and are cancelled by U_4 .

Every term solves the Helmholtz equation exactly at eigenvalue $s(1 + At^2 + Bt^4)^2$. The remaining Dirichlet data are $O(t^5)$ in a fixed smooth norm. Subtract an H^1 boundary lift of that order and test against the normalized first eigenfunction on Ω_t . The pairing has a nonzero limit, as in Section 4, so the eigenvalue error is $O(t^5)$. Analytic dependence from Section 2.1 and block symmetry make the eigenvalue an even function of t , improving the scalar remainder to $O(t^6)$. Taking logarithms proves (8.10). This argument justifies the jet analytically, not just formally.

8.3. Torsion and geometric jets

Write the normalized torsion, volume and perimeter as $1 + X_2 t^2 + X_4 t^4 + O(t^6)$. Their quadratic coefficients are

$$T_2 = \frac{(n+2)(n-3)}{2} m_2, \quad V_2 = \frac{n(n-1)}{2} m_2, \quad P_2 = \frac{n^2 - n + 2}{2} m_2. \quad (8.13)$$

Their quartic coefficients are

$$\begin{aligned} T_4 &= (n+2) \left\{ \left[\frac{n-7}{2} b^2 + \frac{n^2-7n+16}{2} b - \frac{3(n-1)(n-4)}{4} \right] N \right. \\ &\quad \left. + \frac{48+n^3+10n^2-27n}{24} m_4 - \frac{3n(n-1)}{4} m_2^2 \right\}, \\ V_4 &= \frac{n(n-1)}{2} b^2 N + \frac{n(n-1)(n-2)}{2} b N + \frac{n(n-1)(n-2)(n-3)}{24} m_4, \\ P_4 &= \left[\frac{(n-1)(n-2)}{2} + 2(n+2) \right] b^2 N \\ &\quad + \left[\frac{(n-1)(n-2)(n-3)}{2} + 2(n-3)(n+1) \right] b N \\ &\quad + \left[\frac{(n-1)(n-2)(n-3)(n-4)}{24} + \frac{n(n-3)(n-4)}{6} \right] m_4 - \frac{1-8m_2+16m_4}{8}. \end{aligned} \quad (8.14)$$

Here is a derivation of the torsion formula. The first corrections are

$$w_0 = \frac{1-r^2}{2n}, \quad w_1 = \frac{r^2 h}{n}, \quad w_2 = \frac{(b-3/2)r^4 Q - (3/2)m_2}{n}, \quad w_3|_S = \frac{(6-5b)hQ - h^3}{n}. \quad (8.15)$$

They cancel the moving boundary through order three. At order four, harmonic orthogonality gives $\langle h \partial_r w_3 \rangle = 2[(6-5b)N - m_4]/n$, and the mean boundary correction becomes

$$\langle w_4|_S \rangle = \frac{(-7b^2/2 + 8b - 3)N + 2m_4}{n}. \quad (8.16)$$

The fourth-order signed shell integral, divided by $|S^{n-1}|$, is

$$\frac{1}{n} \left\{ \left[\frac{b^2}{2} + \frac{n-7}{2} b + \frac{15-3n}{4} \right] N + \frac{n^2+10n-27}{24} m_4 - \frac{3(n-1)}{4} m_2^2 \right\}. \quad (8.17)$$

For verification, the four contributions before collection, multiplied by $n/|S|$, are $-b^2N/2 - (2n - 1)bN/2 - (n - 1)^2m_4/8$ from w_0 ; $(n + 1)bN + n(n + 1)m_4/6$ from tw_1 ; $(b - 3/2)bN + (b - 3/2)(n + 3)N/2 - 3(n - 1)m_2^2/4$ from t^2w_2 ; and $(6 - 5b)N - m_4$ from t^3w_3 . Adding the ball integral $|S|\langle w_4|_S\rangle/n$ and dividing by $T(B) = |S|/[n^2(n + 2)]$ gives T_4 .

The corrections are finite harmonic polynomials. Their sum has Laplacian exactly -1 and uncanceled boundary value $O(t^5)$. The maximum principle controls the true torsion error. Analyticity and evenness improve the scalar integral remainder to $O(t^6)$.

Volume follows by expanding $\langle R^n \rangle$. For perimeter expand $\langle R^{n-1}\sqrt{1 + |\nabla R|^2/R^2} \rangle$ and use

$$\begin{aligned} \langle |\nabla Q|^2 \rangle &= 4(n + 2)N, & \langle h\nabla h \cdot \nabla Q \rangle &= 2(n + 2)N, & \langle Q|\nabla h|^2 \rangle &= -4N, \\ \langle h^2|\nabla h|^2 \rangle &= \frac{2n}{3}m_4, & \langle |\nabla h|^4 \rangle &= 1 - 8m_2 + 16m_4. \end{aligned} \quad (8.18)$$

These identities follow from (8.2) and integration by parts. In particular the final negative term of P_4 comes from the quartic term in the square-root expansion.

8.4. Complement elimination and an explicit rational coefficient

Use q_V, q_P from (1.2), $\alpha_V = ((n + 2)q_V - 2)/n$ and $\beta_P = ((n + 2)q_P - 2)/(n - 1)$. The two critical quartics are explicitly specified by

$$\begin{aligned} K_V(b) &= 2B - A^2 + q_V(T_4 - T_2^2/2) - \alpha_V(V_4 - V_2^2/2), \\ K_P(b) &= 2B - A^2 + q_P(T_4 - T_2^2/2) - \beta_P(P_4 - P_2^2/2). \end{aligned} \quad (8.19)$$

The coefficient of b^2 is $e_4(q_*)N < 0$ by the established degree-four Hessian. In this balanced symmetry class, the quadratic complement term must be a multiple of Q : products of two degree-two harmonics have only degrees zero, two and four; block exchange excludes degree two here, and the mean-radius normalization excludes zero. The complement equation selects the unique maximum of K_* .

Writing $K_* = u_*b^2 + v_*b + w_*$, the reduced coefficient is therefore

$$b_* = -\frac{v_*}{2u_*}, \quad c_*(n) = w_* - \frac{v_*^2}{4u_*}. \quad (8.20)$$

Higher complement terms do not affect this quartic coefficient because the first velocity belongs to the Hessian kernel. This is the same argument as in Section 6.

To prove nonvanishing without a dimension-by-dimension numerical test, define

$$\begin{aligned} D_V &= -2n^3 - 16n^2 - 24n + (5n + 12)s, \\ D_P &= (15n^2 + 43n + 12)s^2 - (6n^4 + 52n^3 + 116n^2 + 72n)s + 12n^4 + 72n^3 + 96n^2. \end{aligned} \quad (8.21)$$

Rational simplification of (8.5)-(8.20) gives

$$c_V(n) = \frac{N_V(n, s)}{4n^3s(n + 2)^2(n + 6)D_V}, \quad c_P(n) = \frac{N_P(n, s)}{4n^3(n + 2)^2(n + 6)(3n - 1)D_P}. \quad (8.22)$$

Here $N_* = \sum_{r=0}^5 a_{*,r}s^r$, with coefficients

$$\begin{aligned}
a_{V,0} &= -16n^6(n+2)(n+4), \\
a_{V,1} &= 16n^5(n+2)(3n+16), \\
a_{V,2} &= -2n^3(n^4 + 34n^3 + 208n^2 + 136n - 288), \\
a_{V,3} &= n^2(17n^3 + 94n^2 - 512n - 1152), \\
a_{V,4} &= -2n(2n^3 + 3n^2 - 214n - 360), \\
a_{V,5} &= 8(n^2 - 10n - 18),
\end{aligned} \tag{8.23}$$

and

$$\begin{aligned}
a_{P,0} &= -4n^5(n+2)(n+4)(9n^3 - 6n^2 + 49n - 36), \\
a_{P,1} &= 4n^4(n+2)(66n^4 + 269n^3 - 565n^2 - 548n + 108), \\
a_{P,2} &= -n^3(18n^6 + 471n^5 + 1980n^4 - 4895n^3 - 19470n^2 - 8596n + 4608), \\
a_{P,3} &= n^2(117n^5 + 168n^4 - 8229n^3 - 18886n^2 - 4958n + 3960), \\
a_{P,4} &= -2n(3n-1)(6n^4 - 5n^3 - 753n^2 - 1622n - 648), \\
a_{P,5} &= 8(3n-1)(3n^3 - 31n^2 - 68n - 18).
\end{aligned} \tag{8.24}$$

These algebra identities follow by substitution in (8.19) and completing the square as in (8.20).

8.5. Nonvanishing and the critical curve

For each integer $n \geq 2$, both polynomials $N_V(n, s)$ and $N_P(n, s)$ are nonzero rational polynomials in s . Their constant terms in (8.23)-(8.24) suffice: the first is negative, and in the second $9n^3 - 6n^2 + 49n - 36 = n^2(9n - 6) + (49n - 36) > 0$.

The classical Siegel theorem, whose applicable statement is documented in Section 3, makes the nonzero Bessel zero $j_{n/2-1,1}$ transcendental. Its square s is also transcendental. Thus neither numerator in (8.22) can vanish. Neither denominator can vanish: its nonconstant factors are nonzero rational polynomials in s , since $5n + 12 > 0$ and $15n^2 + 43n + 12 > 0$. The intermediate denominator in (8.5) is also nonzero, consistently with spectral simplicity.

Consequently $c_V(n)c_P(n) \neq 0$. Differentiating the reduced energy $e_2(q)m_2t^2 + c_*t^4 + \dots$ and using $\partial_q e_2 \neq 0$ proves (8.4). This completes the balanced nondegeneracy step in Theorem B and Corollary C.

For $n = 2$, $h = \cos 2\theta/2$, $Q = \cos 4\theta/8$. Substituting $t_{\text{planar}} = t/2$ and $b_{\text{planar}} = b/2$ recovers the earlier planar eigenvalue, torsion and perimeter jets exactly. The reduced quartic is one sixteenth of the earlier one, so it is positive. The sign in every higher even dimension is left open.

9. Local versus global maximality

For either F_q or G_q of Section 3, let q_* be its upper local stability threshold. In every dimension $n \geq 2$, there is $\varepsilon_n > 0$ such that for every

$$q_* < q < q_* + \varepsilon_n \tag{9.1}$$

the ball is a strict local maximizer, modulo its Euclidean and scaling symmetries, but is not a global maximizer even among smooth strictly convex domains. The local topology here is the $C^{2,\gamma}$ topology used in the primary source. The constant can differ for the two functionals. No numerical size is asserted.

Proof. The endpoint theorem proved in Sections 3-5 supplies a fixed smooth strictly convex domain Ω_* with

$$\log \frac{F_{q_*}(\Omega_*)}{F_{q_*}(B)} > 0 \quad \text{or} \quad \log \frac{G_{q_*}(\Omega_*)}{G_{q_*}(B)} > 0, \quad (9.2)$$

as appropriate. Choose a sufficiently small but nonzero deformation from that proof, then keep it fixed. For this domain the logarithmic ratio is an affine, hence continuous, function of q . Its strict positivity persists for all q in an interval containing q_* . This proves failure of global optimality in (9.1). Local strict maximality for every $q > q_*$ is already Theorem 1.6 or Theorem 6.3 of Amato-Nitsch-Trombetti-Villone, [final article](#). Combining that established result with (9.2) proves the assertion. This argument does not assert a neighborhood uniform as $q \downarrow q_*$. The competitor is fixed, and need not belong to the shrinking neighborhood on which local maximality holds. $\square$

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